

Sets, Classes, and the First Warnings About Size

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§1.1 Why classes enter before functions

The point of this note is to set up the basic language of sets, classes, indexed families, and functions. The original PDF begins very much in the style of a reading note: it is following the opening sections of a book, but it also pauses at the places where the distinction between a set and a proper class starts to feel less innocent.

Definition 1.1.1. A **set** is a class which can be an element of another class. A class which is not a set is called a **proper class**.

This is not meant to be the whole foundation of set theory, but it explains the working convention of the note: some collections are small enough to be members of other collections, while others are too large and must remain proper classes.

Caution

The original note says that the distinction between sets and proper classes is “not too clear”. Informally this is understandable, but formally this distinction is exactly what prevents the usual paradoxes. In a class theory such as NBG, sets are precisely the classes that may occur as elements of classes; proper classes may not be elements.

§1.2 Russell’s class as the first warning

The first serious example is Russell’s construction. Consider

$$M = \{X \mid X \text{ is a set and } X \notin X\}.$$

At first sight this is a natural class: many sets are not members of themselves. For instance, the set of all books is not itself a book.

Proposition 1.2.1

The class M is not a set.

Proof. Assume, for contradiction, that M is a set. Then it makes sense to ask whether $M \in M$. By the defining condition of M ,

$$M \in M \iff M \notin M,$$

which is impossible. Hence M must be a proper class. $\square$

This is the first place where the note is not merely listing definitions. The example shows why unrestricted comprehension cannot simply produce a set from every condition.

§1.3 Power sets and indexed families

Definition 1.3.1. The **power set axiom** asserts that for every set A , the class

$$\mathcal{P}(A) = \{X \mid X \subseteq A\}$$

of all subsets of A is itself a set. It is also common to write 2^A for this power set.

The next construction is an indexed family. Given a family of classes or sets

$$\{A_i \mid i \in I\},$$

its union and intersection are written

$$\bigcup_{i \in I} A_i = \{x \mid x \in A_i \text{ for some } i \in I\},$$

and

$$\bigcap_{i \in I} A_i = \{x \mid x \in A_i \text{ for all } i \in I\}.$$

Remark 1.3.2 — The original note explicitly points out that the indexing collection I need not be finite or countable. This is a useful mental shift: the notation $\{A_i\}_{i \in I}$ is not just a long list. It may be a genuinely infinite family.

Caution

If I is allowed to be a proper class, then extra care is needed. In ordinary ZFC, one usually speaks of set-indexed families. In a class-theoretic setting, class-indexed families can be discussed, but one must check separately when their unions or intersections are sets.

If A and B are classes, the relative complement of A in B is

$$B \setminus A = \{x \mid x \in B \text{ and } x \notin A\}.$$

If all classes under discussion are subsets of a fixed universe U , then $U \setminus A$ is often denoted A^c .

§1.4 Set identities

The original note lists the standard identities and leaves the proof as easy. A cleaned-up version is:

Proposition 1.4.1

Let A , B , and B_i be subsets of a fixed universe U . Then

$$A \cap \left(\bigcup_{i \in I} B_i \right) = \bigcup_{i \in I} (A \cap B_i),$$

$$A \cup \left(\bigcap_{i \in I} B_i \right) = \bigcap_{i \in I} (A \cup B_i),$$

$$\left(\bigcup_{i \in I} A_i \right)^c = \bigcap_{i \in I} A_i^c, \quad \left(\bigcap_{i \in I} A_i \right)^c = \bigcup_{i \in I} A_i^c.$$

Moreover, $A \cup B = B \cup A$, $A \subseteq B$ means every element of A lies in B , and $A \cap B = A$ when $A \subseteq B$.

Proof. All of these follow by chasing membership. For instance, an element x lies in $A \cap (\bigcup_i B_i)$ exactly when $x \in A$ and $x \in B_i$ for at least one i , which is exactly the condition that $x \in \bigcup_i (A \cap B_i)$. The De Morgan laws are proved similarly after replacing “there exists” by “for all” under negation. $\square$

§1.5 Images, inverse images, and functions

Let $f : A \rightarrow B$ be a function. If $S \subseteq A$, the image of S under f is

$$f(S) = \{b \in B \mid b = f(a) \text{ for some } a \in S\}.$$

If $T \subseteq B$, the inverse image of T is

$$f^{-1}(T) = \{a \in A \mid f(a) \in T\}.$$

Example 1.5.1

For $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x^2$, we have

$$f(\mathbb{R}) = [0, \infty), \quad f^{-1}(\{9\}) = \{-3, 3\}.$$

A map $f : A \rightarrow B$ is **injective** if $f(a) = f(a')$ implies $a = a'$. It is **surjective** if $f(A) = B$. It is **bijective** if it is both injective and surjective.

Proposition 1.5.2

For composable maps $A \xrightarrow{f} B \xrightarrow{g} C$:

- (i) if f and g are injective, then $g \circ f$ is injective;
- (ii) if f and g are surjective, then $g \circ f$ is surjective;
- (iii) if $g \circ f$ is injective, then f is injective;
- (iv) if $g \circ f$ is surjective, then g is surjective.

§1.6 One-sided inverses

The last theorem in the original note is about recognizing injective and surjective maps through one-sided inverses.

Theorem 1.6.1

Let $f : A \rightarrow B$ be a function and assume $A \neq \emptyset$.

- (i) f is injective if and only if there is a map $g : B \rightarrow A$ such that $g \circ f = 1_A$.
- (ii) f is surjective if and only if there is a map $h : B \rightarrow A$ such that $f \circ h = 1_B$, provided the required choices of elements from the fibres of f can be made.

Proof. For (i), suppose f is injective. For $b \in f(A)$, there is a unique $a \in A$ with $f(a) = b$. Choose once and for all an element $a_0 \in A$. Define

$$g(b) = \begin{cases} a, & b \in f(A) \text{ and } f(a) = b, \\ a_0, & b \notin f(A). \end{cases}$$

Then $g(f(a)) = a$ for all $a \in A$. Conversely, if $g \circ f = 1_A$, then $f(a) = f(a')$ implies

$$a = g(f(a)) = g(f(a')) = a',$$

so f is injective.

For (ii), if h exists and $f \circ h = 1_B$, then every $b \in B$ equals $f(h(b))$, so f is surjective. Conversely, if f is surjective, each fibre $f^{-1}(b)$ is nonempty. Choosing $h(b) \in f^{-1}(b)$ gives a right inverse. $\square$

Caution

The last choice of one element from each fibre is exactly where the Axiom of Choice may enter. The original note itself flags this point in a footnote, and it will reappear in the later note on choice and cardinal numbers.

§1.7 What remains from this note

The route is simple but important: naive collections lead to Russell's class; power sets and indexed families give the first working constructions; functions are then organized by images, inverse images, and one-sided inverses. The main gap is not computational but foundational: one has to keep track of when a class is actually a set.